CSET485 – AI AND SOCIETY

ASSIGNMENT #1

Predictive Modelling, Statistical Stress-Testing and Fairness Trade-Off Analysis

Student	Amit Tiwari
Enrollment	S24CSEU0627
Assigned Dataset	Adult Census Income Dataset
Random State	627
Train/Test Split	70% / 30%

Prepared for CSET485 – AI and Society

Submission note: the Jupyter notebook is supplied separately. Run all cells from a fresh kernel before submission; downloading the UCI Adult dataset requires internet access.

Executive Summary

This assignment examines income classification using the UCI Adult Census Income dataset. The notebook combines the source training and test files into 48,842 rows and 16 columns, creates a binary target for income above $50K, and uses a stratified 70:30 split with random_state = 627. The analysis compares Linear Regression thresholded at the median test prediction with Logistic Regression using a 0.50 probability threshold, then examines subgroup coeficients, model-specification sensitivity, high-confidence errors, and fairness ratios.

Measure	Reported result
Dataset rows / columns	48,842 / 16
Training / test rows	34,189 / 14,653
Logistic accuracy	0.8543
Logistic precision / recall / F1	0.7307 / 0.6192 / 0.6704
Logistic ROC-AUC	0.9057
OLS vs logistic disagreements	4,382 of 14,653 (29.91%)
High-confidence errors identified	725 (4.948% of test set); 50 sampled

1. Data Preprocessing

The Adult dataset is combined from the UCI source files. The target income_binary is 1 for income above $50K and 0 for income at or below $50K. The class counts are 37,155 at or below $50K and 11,687 above $50K. The split contans 34,189 training rows and 14,653 test rows; the positive-class rate is 0.2393 in both sets. The source uses “?” to mark missing categorical values. Missing values include occupation (2,809), workclass (2,799), and native_country (857).

Predictors, preprocessing and leakage control

Numeric features are imputed using the median of the training set and then standardized. Categorical features are imputed by the most frequent value in the training set and then one-hot encoded. Test-set unknown categories are ignored. This pipeline is trained on the traning set to prevent test set lekage. Features include age, fnlwgt, education_num, capital_gain, capital_loss, hours_per_week, workclass, marital_status, occupation, relationship, race, sex, and native_country. Education feature is dropped since education_num represents ordered education level.

Numeric variable	IQR-flagged observations	Share
age	216	0.44%
fnlwgt	1,453	2.97%
education_num	1,794	3.67%
capital_gain	4,035	8.26%
capital_loss	2,282	4.67%
hours_per_week	13,496	27.63%

IQR flags are diagnostic, not proof of invalid data. Capital gain and capital loss have quartiles concentrated at zero, so their upper bounds are zero and many non-zero observations are flagged. The notebook retains these observations rather than automatically deleting potentially meaningful data.

2. Part 1 — Build and Compare Models

Model A — Linear Regression

Ordinary least-squares Linear Regression is fitted to the binary incme target. Predictins are classified as high when they are at least the median test prediction, 0.222977. OLS predictions range from −0.4125 to 1.7520, showing that OLS outputs are not constrained to the probability interval [0, 1].

Model B — Logistic Regression

Logistic Regression estimates the probability of the binary target and labels a case positive when its probability is at least 0.50. It achieves accuracy 0.8543, precision 0.7307, recall 0.6192, F1-score 0.6704 and ROC-AUC 0.9057.

Agreement matrix	Logistic: Low	Logistic: High
OLS: Low	7,313	13
OLS: High	4,369	2,958

The methods disagree on 4,382 of 14,653 cases (29.91%) and agree on 10,271 cases (70.09%). The methods optimize different objectives and use different cut-offs: OLS minimizes squared error and uses a data-dependent median score; logistic regression estimates a binary probability and applies 0.50. These thresholds are not interchangeable.

contain 48,842 rows and 16 columns. The target variable income_binary is equal to 1 if the income is greater than $50K and 0 if the income is less than or equal to $50K. The count of the two classes is 37,155 for less than or equal to $50K and 11,687 for greater than $50K. The split consists of 34,189 training examples and 14,653 testing examples using a stratified split of 70:30 and a random state of 627. The positive class ratio is 0.2393 in both datasets. The source dataset utilizes '?' to represent missing categorical values. According to the notebook, there are 2,809 missing occupations, 2,799 workclasses, and 857 native_country; all other columns in the displayed dataframe have zero missing values after reading the data. All numeric features are imputed with the median value from the training set and scaled. All categorical features are imputed using the most frequent value from the training set and are one-hot encoded, while unseen test categories are ignored. The predictors are age, fnlwgt, education_num, capital_gain, capital_loss, hours_per_week,

workclass, marital_status, occupation, relationship, race, sex, and native_country. The predictor "text" education has not been included in the feature set since education_num represents ordered levels of education. The outlier test conducted in the notebook applies the IQR rule to identify unusual values in numeric features. This test identified 216 outliers in age (0.44%), 1,453 in fnlwgt (2.97%), 1,794 in education_num (3.67%), 4,035 in capital_gain (8.26%), 2,282 in capital_loss (4.67%) and 13,496 in hours_per_week (27.63%). The IQR upper bound for capital gain and capital loss is zero since the quartiles for both features concentrate at zero. Hence, many non-zero values are flagged by this rule. These are only indicators but not proof that the data is invalid.

Linear Regression trained on the binary income target. It is considered a high prediction if it is at least the median test prediction. Model B is Logistic Regression and is considered a high prediction if the probability is at least 0.5. Median test prediction for OLS is 0.222977. OLS predictions vary from -0.4125 to 1.7520, showing that OLS predictions are not limited to probability values ranging from 0 to 1. Agreement matrix Logistic low (0) Logistic high (1) OLS low (0) 7,313 OLS high (1) 4,369 2,958 The models have disagreement in 4,382 cases out of 14,653 cases (29.91%). The disagreements are asymmetric as 4,369 cases have OLS-high/logistic-low disagreement, while there are only 13 cases of OLS-low/logistic-high disagreement. OLS accuracy against the true labels after applying threshold is 0.7126. Logistic Regression has the following performance: accuracy 0.8543, precision 0.7307, recall 0.6192, F1-score 0.6704, and ROC-AUC 0.9057. The differences come from the fact that OLS tries to minimize square error, and the cutoff is data-dependent median score. On the other hand, Logistic Regression is used to estimate probabilities with the help of logistic link and fixed 0.5 cutoff. The median OLS score is not equivalent to calibrated 50% probability.

2.1 Simpson’s paradox hunt: subgroup coefficients

The notebook fits logistic regression to the the entire dataset and to the Female and Male subgroups individually and compares the coefficient for age and its 95% confidence interval. Model Age coefficient 95% confidence interval Displayed p-value Full dataset 0.02275 [0.02033, 0.02517] 0.0 (rounded) Model Age coefficient 95% confidence interval Displayed p-value Female subgroup 0.02895 [0.02287, 0.03504] 0.0 (rounded) Male subgroup 0.02139 [0.01876, 0.02403] 0.0 (rounded) In all three cases, the coefficient is positive, meaning that this run does not provide an example of a sign reversal and thus cannot be called a verified case of Simpson’s paradox. What is clear is the difference in size of the coefficients: the coefficient for age in the Female subgroup (0.02895) is larger than the pooled sample coefficient (0.02275) and the coefficient for age in the Male subgroup (0.02139). These coefficients describe conditional changes in the log-odds and not causal effects. The p-values are shown as 0.0 due to rounding; they can thus be described as very small at the displayed level of precision, but not literally equal to zero. A policy recommendation based on the pooled effect might ignore the larger conditional association in the Female subgroup. Nevertheless, confidence intervals alone do not confirm the difference between subgroup coefficients;

2.2 Omitted-variable bias: removing education_num

Below are the four coefficient comparisons as described in the notebook. Percentage changes are the percentage changes as defined in the notebook relative to the absolute full model coefficient. Predictor Full coefficient Reduced coefficient Change (reduced − full) Absolute % change age 0.02275 0.01922 −0.00353 15.52% hours_per_week 0.03022 0.03246 +0.00224 7.42% capital_gain 0.00032 0.00032 +0.00001 2.06% capital_loss 0.00066 0.00073 +0.00007 11.05% The greatest absolute percent change is for age at 15.52%, reducing from 0.02275 to 0.01922. There is an increase of 7.42% in hours_per_week, 2.06% change in capital_gain, and an increase of 11.05% in capital_loss. Such sensitivity implies that the coefficient estimates will depend on the set of variables used in the model. Age and level of education have some similarities when it comes to association with income. The exclusion of education_num can lead to changes in the proportion of association between other predictors and the response. This experiment is intended as a stress test and not conclusive evidence that education_num is the only confounding variable or that any coefficient is causal.

2.3 Adversarial subset construction

The notebook identifies 725 high-confidence incorrect predictions before sampling, equivalent to 4.948% of the total 14,653 rows in the test set. It randomly samples 50 rows to compare demographics. The sampled set has 39 high-confidence false negatives (income above $50K and the predicted probability <= 0.25) and 11 high-confidence false positives (income <= $50K and the predicted probability >= 0.75). Characteristic Adversarial subset (n=50) Full test set (n=14,653) Average age 46.080 38.571 Male 72.00% 67.11% Female 28.00% 32.89% White 88.00% 86.20% Black 12.00% 9.23% Married-civ-spouse 58.00% 46.04% Husband 52.00% 40.65% Not-in-family 32.00% 25.89% The two most obvious over-represented categories in this sample are Married-civ-spouse (58%, vs 46.04% in the test set) and Husband (52%, vs 40.65%).The sample is also older on average by about 7.51 years (46.080 versus 38.571). These are descriptive contrasts, not evidence that marriage, sex, race, or age causes model failure. These 50 rows have been drawn out of a total of 725 potential rows, hence percentages in the sample may differ; conclusions regarding similarity should be made carefully. The reason for this might be in the fact that there could be some non-linear relationship between the variables age, education, marital status, occupation, and working hours which is not covered by the additive logistic model. The target variable is also somewhat vague, given the available predictors, as other significant factors affecting income are not included.

2.4 Fairness trade-off analysis

The sensitive attribute is sex: Group A = Female and Group B = Male . The notebook defines DP as Female selection rate divided by Male selection rate; EO as the minimum of the Female/Male TPR ratio and Female/Male FPR ratio; and PP as Female PPV divided by Male PPV. A ratio of 1 represents parity under that metric. Threshold DP ratio EO ratio PP ratio Accuracy 0.05 0.4708 0.4480 0.6784 0.6287 0.15 0.3042 0.2561 0.9130 0.7551 0.25 0.2897 0.2414 0.9432 0.8090 0.35 0.2937 0.2533 0.9370 0.8382 0.45 0.2877 0.2431 0.9620
0.8514 0.55 0.2648 0.2247 0.9694 0.8525 0.65 0.2635 0.2217 0.9802 0.8452 0.75 0.2731 0.2501 0.9720 0.8277 0.85 0.2847 0.3215 0.9594 0.8073 0.95 0.2728 0.2258 0.9987 0.7886 Figure 1. DP-versus-EO trade-off across thresholds. No threshold makes DP, EO, and PP all equal to 1. At threshold 0.35, DP = 0.2937, EO = 0.2533, PP = 0.9370, and accuracy = 0.8382. The Female/Male selection rates are 0.1079 and 0.3674; TPRs are 0.6260 and 0.7721; FPRs are 0.0479 and 0.1892; PPVs are 0.6019 and 0.6424. Threshold 0.55 gives the highest listed accuracy (0.8525), but DP and EO fall to 0.2648 and 0.2247. Threshold 0.95 gives near parity in PP (0.9987), but accuracy falls to 0.7886 and DP/EO remain far from 1. The observed table does not support a claim that improving DP necessarily causes EO to fall: both ratios vary non-monotonically across thresholds. Moving from threshold 0.05 to 0.55 changes DP from 0.4708 to 0.2648 (a 43.8% relative decrease, farther from parity) and EO from 0.4480 to 0.2247 (a 49.8% relative decrease, also farther from parity). This is not a DP-improvement/EO-loss example. Interpret the actual values rather than inventing a trade-off direction.

4. Part 3 — Descriptive analysis questions

4. Part 3 — Descriptive Analysis Questions

Q1. Why did the threshold-based approaches flag different individuals?

The disagreement between the two methods occurs for 4,382 of 14,653 observations, i.e., 29.91% of the test data. The agreement matrix between the two models shows that there are 7,313 low classifications in common between OLS and logistic regression as well as 2,958 high classifications in common between the two models. On the other hand, there are 4,369 high classifications by OLS and low classifications by logistic regression, but there are only 13 low classifications by OLS and high classifications by logistic regression. The median of OLS predictions is 0.222977, which is significantly lower than the 0.5 threshold used for logistic probabilities. Nevertheless, these two cut-offs cannot be considered equal to each other since OLS attempts to minimize the squared error on the binary target variable and may yield predictions outside the [0, 1] interval; in this run its predictions vary from −0.4125 to 1.7520. The median cut-off is defined by the distribution of OLS scores and splits relatively in the middle of them. As seen in the Asymmetric Disagreement Matrix, the median OLS rule marked considerably higher numbers of cases to be high compared to the logistic rule. Compared with the ground truth, the OLS threshold accuracy is 0.7126, while the logistic accuracy is 0.8543; the logistic precision is 0.7307, recall 0.6192, F1 0.6704, and ROC-AUC 0.9057. All the above measures recommend Logistic Regression in this binary case for the current split. However, as seen from recall, quite many high-income examples are missing. The main point here is that the goals of model, score interpretation, and thresholds influence the decisions. Neither of these thresholds is neutral; classification differences must be analyzed before employing the score.

Q2. What did the subgroup analysis reveal?

The full-model age coefficient is 0.02275 with a 95% confidence interval of [0.02033, 0.02517]. In the Female subgroup it is 0.02895 [0.02287, 0.03504], and in the Male subgroup it is 0.02139 [0.01876, 0.02403]. The age coefficient is positive in all three fits, so the output does not demonstrate a sign reversal and should not be presented as a confirmed Simpson’s paradox. Instead, the coefficient is larger in the Female subgroup than in the Male subgroup and the pooled model. In logistic regression, a coefficient represents a conditional change in log-odds associated with a one-unit predictor increase while holding includd predictors fixed; it is not a causal effect. The notebook displays p-values as 0.0 because of rounding, so the report should say they are very small at the displayed precision rather than exactly zero. The subgroup comparison suggests that the pooled modl may conceal differences in the age-income association. If a policymaker only saw the pooled coefficient, they might assume the relationship is equally strong across demographic groups and miss a potentially important difference in how the model represents income. Still, coefficient magnitude alone does not prove a statistically significant difference between groups. The confidence intervals overlap to some degree, and a formal sex-by-age interaction test would provide a more direct test of effect heterogeneity. The responsible conclusion is that subgroup effects differ in magnitude in this run and warrant furter testing. Policy decisions should therefore consider subgroup results and uncertainty, rather than treating the full-sample coefficient as universally representative.

Q3. What changed when education_num was omitted?

The omitted-variable test removes education_num from the reduced logistic model and compares four remaining coefficients. Age decreases from 0.02275 to 0.01922, a change of −0.00353 and an absolute percentage shift of 15.52%. Hours_per_week increases from 0.03022 to 0.03246, a change of +0.00224 or 7.42%. Capital_gain is reported as 0.00032 in both rounded columns, but the underlying notebook calculation gives a 2.06% absolute percentage shift. Capital_loss increases from 0.00066 to 0.00073, a change of +0.00007 or 11.05%. Among the four predictors, age has the largest absolute percentage change. This result illustrates that model coefficients depend on specification: when an income-related predictor is removed, other coefficients can absorb some of the association previously represented by that variable. Age and educationl attainent may contain overlapping information about employment and earnings, which is consistent with the shift in the age coefficient. It is important not to overstate the result. The experiment does not prove that education_num is a causal confounder, nor that the reduced model is always invalid. Omitted-variable bias depends on the relationships among the omitted predictor, included predictors, and outcome, and the model is observational. In practice, analysts should explain why variables are included or excluded, compare plausible specifications, consult subject-matter experts, inspect subgroup and residual patterns, and validate conclusions on external data. Reporting only one model specification would conceal this sensitivity. The coefficient table is therfore evidence for caution and documentation, not a causal estimate.

Q4. How would a practitioner detect a hidden confound that cannot currently be observed?

I would use a structured domain-expert audit combined with external validation and sensitivity analysis. First, interview people familiar with the process that generated the dataset—for example, labor economists, survey researchers, and practitioners who understand income measurement. Ask what variables could affect both the predictors and income but are missing from the data, such as local labor-market conditions, industry, household responsibilities, regional costs, or changes in survey practice. Document the proposed relationships in a causal diagram and distinguish assumptions from measured evidence. Second, test the model on a separate time period or region. If calibration, error rates, or subgroup patterns change substntially, the modl may rely on relationships that do not generalize. Third, inspect residuals and errors across age, sex, race, education, occupation, and marital-status groups, then test plausible nonlinear terms and interactions. The current notebook gives a useful lead: the 50 sampled confident errors have mean age 46.080, compared with 38.571 in the full test set, and 58% are Married-civ-spouse compared with 46.04% in the test set. These differences do not prove confounding, but they identify patterns worth investigating. Finally, perform sensitivity analysis to estimate how strongly a hypothetical unmeasured variable would need to relate to predictors and outcome to change the conclusions. No one method can guarantee detetion of a completely unobserved confound. The goal is a documented process that makes assumptions visible, checks generalization, and updats the model when credible new evidence becomes available.

Q5. What characteristics appear in the 50-row adversarial subset, and why might the model fail?

The notebook identifies 725 high-confidence incorrect predictions in the full test set, 4.948% of the 14,653 test cases, and randomly samples 50 of them. Within this sample, 39 are high-confidence false negatives and 11 are high-confidence false positives. The most noticeable over-represented categories are Married-civ-spouse, which accounts for 58% of the sample compared with 46.04% of the test set, and Husband relationship, which accounts for 52% compared with 40.65%. The mean age is 46.080 years in the subset versus 38.571 years in the full test set. Men represent 72% of the sample versus 67.11% of the test set. These percentages describe the sample; they do not prove that any demographic trait causes error. Since the notbook samples 50 cases from 725 eligible cases, the exact percentages could change with a different sample. A plausible statistical explanation is that the logistic model uses additive feature effects and may fail to represent nonlinear patterns or interactions, such as age having different associations at different education levels or occupations. Other variables that influnce income may not be recorded, and two people with similar observed features can have different outcomes. A model may be confidently wrong when its learned probability is poorly calibrated for a particular combination of features or when that combination is sparsely represented in training data. The correct next step is to inspect all 725 confident errors, calculate error rates across subgrups, and test interaction terms and calibration.

The sample identifies a diagnostic direction, not a universal causal explanation.

Q6. What does the fairness trade-off mean for a real person?

Consider a hypothetical income-classification tool used to decide who receives additional review for a financial-support program. At threshold 0.35, the notebook reports a positive-selection rate of 0.1079 for Female records and 0.3674 for Male records, producing a DP ratio of 0.2937. The true-positive rates are 0.6260 for Female and 0.7721 for Male records, while false-positive rates are 0.0479 and 0.1892. Imagine two people with model probabilities just above 0.35, one Female and one Male. Both would be classfied positive at that threshold, but the group-level rates show that far fewer Female records are selected overall. If the threshold is raised to 0.55, the model's accuracy increases from 0.8382 to 0.8525, yet DP falls from 0.2937 to 0.2648 and EO from 0.2533 to 0.2247. That change means an accuracy-oriented choice can worsen parity ratios and can change who recives a positive classification. If the tool were used for screening, a false negative might mean a person who needs support is not reviewed; a false positive might consume scarce review capacity or trigger unnecessary scrutiny. The consequences depend on the actual use, which this dataset does not specify. Fairness metrics are group-level summaries, so they cannot determine the outcome for a particular person or establish that the model should be used. A responsble practitioner should make the threshold rationale explicit, test error rates for affected groups, provide human review and appeal, and assess whether historical income labels are an appropriate basis for the decision at all.

Q7. Which threshold is defensible, and what is the strongest counterargument?

Threshold 0.35 is a candidate for discussion, not a uniquely optimal answer. In the notebook it produces accuracy 0.8382, DP ratio 0.2937, EO ratio 0.2533, and PP ratio 0.9370. The Female/Male selection rates are 0.1079 and 0.3674; their TPRs are 0.6260 and 0.7721, respectively. These figures provide a concrete balance to discuss: accuracy is reasonably high in aggregate and PP is closer to 1 than DP or EO, but selection and error-rate ratios remain far from parity. Threshold 0.55 offers the highest listed accuray, 0.8525, but DP drops to 0.2648 and EO to 0.2247. Threshold 0.95 brings PP to 0.9987, yet accuracy declines to 0.7886 and DP/EO remain far from 1. There is no threshold in the sweep that achieves parity across all three metrics. A critic's strongest counterargument is that choosing 0.35 still accepts a DP ratio of only 0.2937, meaning the Female positive-selection rate is much lower than the Male rate under the assignment's definition. That objection is vald and prevents describing the threshold as simply fair. My position is that 0.35 can be used as an illustrative compromise for further evaluation only if the application's harms, fairness priorities, and safeguards are specified. It should not be deployed operationally based on this table alone. Threshold selection is a normative decision informed by statistical evidence, not something accuracy alone can settle.

Q8. If forced to choose, should a policymaker prioritize demographic parity or equalized odds?

I would not choose a fairness metric without first clarifying the decision's purpose and the harms of different errors. If the policymaker nevertheless required one priority for an eligibility-screening scenario, I would lean toward equalized odds as an initial safeguard against groups experiencing very different true-positive and false-positive rates, while explicitly monitoring demographic parity as a separate concern. At threshold 0.35, DP is 0.2937 and EO is 0.2533; at threshold 0.55, DP is 0.2648 and EO is 0.2247. At threshold 0.85, EO reaches its highest listed value of 0.3215, while DP is 0.2847, PP is 0.9594, and accuracy is 0.8073. None of these values is close to perfect parity for DP or EO. Prioritizing equalized odds would therefore not solve the selection disparity: the DP ratio remains far below 1. In some settings, demographic parity may be especily important because unequal selection can reproduce historical exclusion; that is the principal argument against my preference. The acceptable trade-off depends on whether the system is allocating a benefit, identifying a risk, or merely deciding which cases need human review. The Adult dataset records historical income and does not validate this model for any real eligibility decision. I would therefore not recommend high-impact deployment based on these results. The responsible polcy is to specify the fairness objective with affected stakeholders, disclose the metric sacrificed, examine alternative models and thresholds, validate externally, and establish review and appeal mechanisms.

5. Part 4 — Reflection: Should the Model Be Deployed?

Direct answer. I would not recommend deploying this model as a standalone decision-maker in a high-impact setting. The model has useful predictive signal, but the stress tests show that aggregate performance does not capture subgroup differences, confident errors, model-specification sensitivity, or fairness trade-offs. A single number fails first because aggregate performance hides different types of mistakes. Logistic Regression reaches accuracy 0.8543 and ROC-AUC 0.9057, which might look strong in a summary. However, recall is 0.6192, so a substantial share of actual above-$50K cases are missed at the 0.5 threshold. The OLS-based method reches accuracy 0.7126 and disagrees with logistic classification on 29.91% of the test set. These results show that the headline score depends on the model and threshold

and does not describe who experiences false positives or false negatives. Second, threshold selection is partly a values decision. At threshold 0.35, accuracy is 0.8382, DP is 0.2937, EO is 0.2533, and PP is 0.9370. At threshold 0.55, accuracy rises to 0.8525, while DP and EO fall to 0.2648 and 0.2247. At threshold 0.95, PP reaches 0.9987, but accuracy drops to 0.7886. No threshold achieves parity on all three ratios. The data therefore do not identify one uniquely fair cutoff; decision-makers must explain which errors and disparities matter most. Third, confident mistakes occur in a meaningful subset. The notebook finds 725 high-confidence wrong predictions, 4.948% of the test set. Among 50 sampled cases, 39 are false negatives and 11 are false positves. The sample is older on average (46.080 years) than the full test set (38.571), and 58% are Married-civ-spouse compared with 46.04% of the test set. These patterns merit error analysis but do not prove that demographic traits cause errors. Fourth, coefficient estimates change with model specification. Omitting education_num shifts the age coefficient from 0.02275 to 0.01922, an absolute percentage change of 15.52%. This demonstrates sensitivity, not causality; hidden variables and distribution shifts may remain. The subgroup analysis also found age coefficients of 0.02895 for Female records and 0.02139 for Male records, compared with 0.02275 overall. Their signs do not reverse, but the difference in magnitude is a reason to test interactions and report uncertainty rather than relying only on the pooled estimate. Before considering any deployment, I would require external validation, calibration checks, subgroup monitring, documented threshold rationale, human review, and an appeal mechanism. I would also ask whether income above $50K is an appropriate target for the proposed decision and whether affected people have meanngful recourse. The Adult dataset reflects historical income patterns; it should not become a moral or eligibility score. Responsible AI means explaining uncertainty and trade-offs rather than reducing a consequential decision to one performance number.

6. Conclusion

changes classifications, subgroup age coefficients differ in magnitude, omitting education_num changes other coefficients, and confident errors occur in a subset whose demographic composition differs from the test set. Across the ten tested thresholds, none achieves parity across DP, EO, and PP. These findings

support cautious interpretation and further validation, not standalone deployment. Numerical values and tables in this report were checked against the saved executed notebook outputs; because the runtime could not access the UCI download host during this review, the notbook was not rerun from a clean session. The report therefore claims output-to-report consistency, not an independent clean-run reproduction.

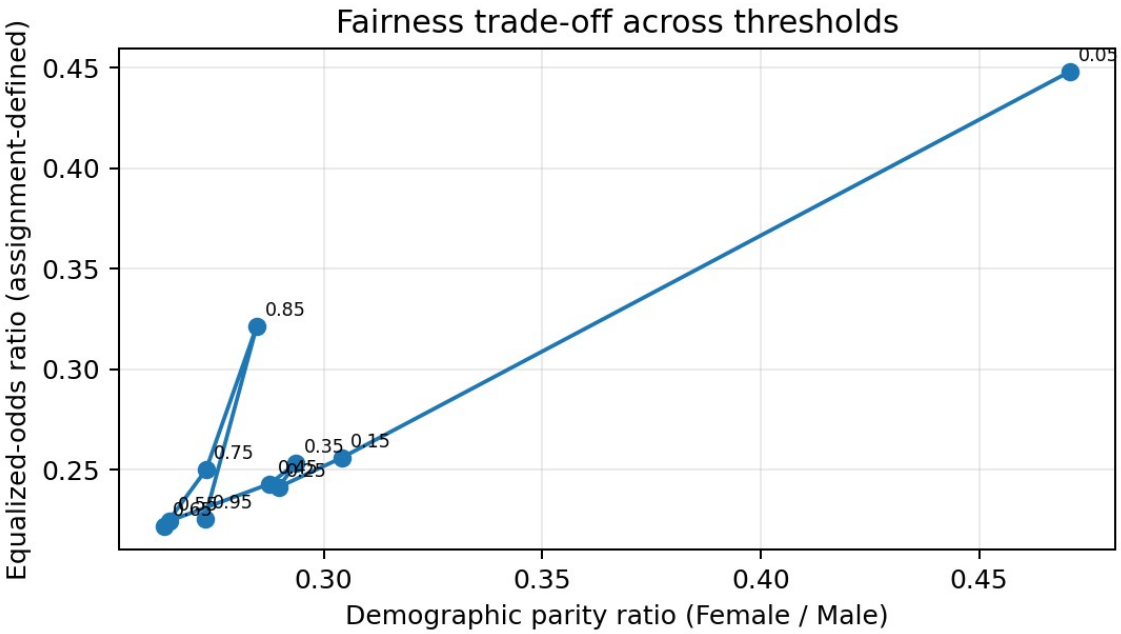
References

1. UCI Machine Learning Repository, Adult dataset: https://archive.ics.uci.edu/ml/machine-learning-databases/adult/ 2. CSET485 – AI and Society, Assignment 1 and Assignment Evaluation Rubrics supplied with the course.

Appendix A — Fairness Threshold Sweep

The notebook defines DP as the Female/Male selection-rate ratio, EO as the minimum of Female/Male TPR and FPR ratios, and PP as the Female/Male PPV ratio. Values near 1 indicate parity for that particular ratio; these measures should not be treated as interchangeable or as a complete definition of fairness.

Threshold	DP ratio	EO ratio	PP ratio	Accuracy
0.05	0.4708	0.4480	0.6784	0.6287
0.15	0.3042	0.2561	0.9130	0.7551
0.25	0.2897	0.2414	0.9432	0.8090
0.35	0.2937	0.2533	0.9370	0.8382
0.45	0.2877	0.2431	0.9620	0.8514
0.55	0.2648	0.2247	0.9694	0.8525
0.65	0.2635	0.2217	0.9802	0.8452
0.75	0.2731	0.2501	0.9720	0.8277
0.85	0.2847	0.3215	0.9594	0.8073
0.95	0.2728	0.2258	0.9987	0.7886



Interpretation: no tested threshold makes DP, EO and PP all equal to 1. At threshold 0.35, accuracy is 0.8382, DP is 0.2937, EO is 0.2533 and PP is 0.9370. Threshold 0.55 has the highest listed accuracy (0.8525), but DP and EO are lower. These results illustrate that improving aggregate accuracy does not guarantee better parity. The EO ratio is the assignment-defined minimum summary; inspect the underlying TPR and FPR separately.

Appendix B — Reproducibility Checklist

• Enrollment-based random_state is 627, matching S24CSEU0627.

• Stratified train/test split is 70/30.

• Imputation, scaling and encoding are fitted within pipelines on training data.

• Outlier analysis is diagnostic; flagged observations are retained.

- OLS and logistic-regression comparison includes the agreement matrix and performance metrics.
- Subgroup coefficient analysis, omitted-variable sensitivity, high-confidence error subset and fairness threshold sweepare included.
- Restart the kernel and run all cells before submission; UCI data download requires internet access.
- Ensure the report and notebook contain the same final outputs and upload both files.